

Personal Firewall Security Dashboard

USER MANUAL

VERSION 1.0



Developed by **Shivansh** · Cyber Security Internship Project

1. Introduction

The **Personal Firewall Security Dashboard** is a Python-based cybersecurity application designed to monitor active network connections and manage firewall rules through a graphical user interface.

The application allows users to monitor live network activity, manage blocked IP addresses and ports, receive security alerts, and maintain event logs. The project serves as a practical implementation of firewall management concepts and network monitoring techniques.

Network Monitoring

Monitor live network activity in real time

Firewall Rules

Manage blocked IP addresses and ports

Security Alerts

Receive alerts from the monitoring system

Event Logs

Maintain detailed system event logs

2. System Requirements

Hardware Requirements

- Intel i3 Processor or higher
- 4 GB RAM or above
- 500 MB Free Storage

Software Requirements

- Windows 10 / Windows 11
- Python 3.10 or higher
- Visual Studio Code (Optional)
- Internet Connection

Python Dependencies

psutil

System and process utilities



Install required dependencies using: `pip install -r requirements.txt`

3. Installation Guide

01

Download the Project

Clone the repository or download the project files.

```
git clone <repository-link>
```

03

Install Dependencies

```
pip install -r requirements.txt
```

02

Navigate to Project Directory

```
cd Personal-Firewall
```

04

Verify Project Structure

Ensure the following folders exist:

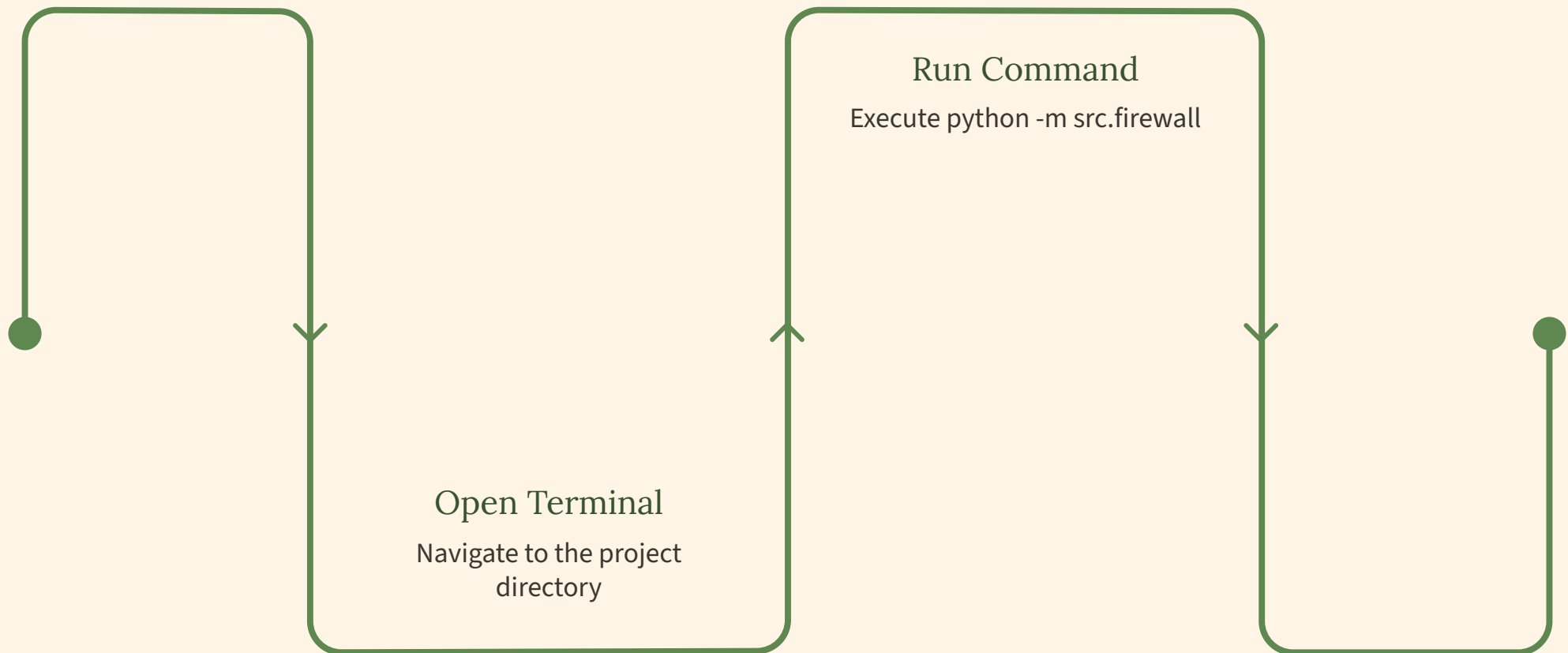
```
src/ config/ logs/ tests/ screenshots/ docs/
```

4. Running the Application

Open a terminal inside the project directory and execute the following command:

```
python -m src.firewall
```

The application dashboard will open automatically.



5. Dashboard Overview

The dashboard contains the following sections:

Status Section

Displays current firewall status.

Example: STATUS: ACTIVE

Statistics Panel

Shows:

- Active Connections
- Total Rules
- Alert Count
- System Uptime

Blocked IP Panel

Displays all blocked IP addresses configured in the firewall.

Blocked Port Panel

Displays all blocked ports configured in the firewall.

Live Connections Panel

Shows active network connections currently detected on the system.

Security Alerts Panel

Displays alerts generated by the firewall monitoring system.

6. Managing Firewall Rules

Adding a Blocked IP Address

- 1 Click the "Add IP" button
- 2 Enter a valid IP address
- 3 Click OK
- 4 The IP address will appear in the Blocked IP list

 Example: 192.168.1.100

Removing a Blocked IP Address

- 1 Select an IP address from the list
- 2 Click the "Remove IP" button
- 3 The selected IP address will be removed

6. Managing Firewall Rules (Continued)

Adding a Blocked Port

- 1 Click the "Add Port" button
- 2 Enter a valid port number
- 3 Click OK
- 4 The port will be added to the Blocked Port list

 Example: 443

Removing a Blocked Port

- 1 Select a port from the list
- 2 Click the "Remove Port" button
- 3 The selected port will be removed

7. Security Alerts

The Security Alerts panel displays real-time alerts generated by the firewall monitoring system.



Monitoring

Alerts are generated automatically by the firewall monitoring system.



Display

All alerts are displayed in the Security Alerts panel.



Real-Time

Alerts are updated in real time as the system monitors network activity.



8. Project Outcome

The Personal Firewall Security Dashboard demonstrates core cybersecurity concepts including:

Network Monitoring

Real-time tracking of active connections

Access Control

IP and port blocking capabilities

Threat Detection

Automated security alert generation

Event Logging

Comprehensive system activity records

This project serves as a practical implementation of firewall management concepts and network monitoring techniques.



For questions or support, contact the developer: **Shivansh (shivanshsh213@gmail.com)**